PODDAR
PIPES

PRESSURE PIPES FOR **WATER SUPPLY**

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PRESSURE PIPES FOR WATER SUPPLY

LEAD-FREE uPVC PRESSURE PIPES FOR AGRICULTURAL **WATER SUPPLY**

Poddar Pipes manufactures a completely lead-free Agri solvent-weld uPVC pressure pipe and fittings system, carefully engineered to meet the demands of modern agricultural water supply.

Designed for durability, UV resistance, safety, and consistent performance, this system offers farmers and agricultural professionals a dependable and cost-effective solution for conveying clean, potable water across their fields and irrigation networks.

UV-RESISTANT

LEAD-FREE

CORROSION RESISTANCE

A LEGACY OF QUALITY

ABOUT
PODDAR PIPES

Poddar Pipes are engineered to strengthen and elevate infrastructure of every scale, delivering solid and dependable foundations even for the most complex and demanding project requirements. Our forward-thinking piping solutions have earned the trust of engineers, builders, and architects across the industry, who rely on them to ensure seamless system integration and consistently optimal performance over the long term. When you choose Poddar, you invest in pipes that are built to reinforce your vision, support your goals, and take your project to the next level.

STANDARDS WE STAND BY

OUR COMMITMENT TO **QUALITY CERTIFICATIONS**

OUR COMMITMENT TO YOU

BUILT ON TRUST AND **ASSURANCES**

Every pipe and fitting that leaves our facility carries with it a promise — a promise of quality, precision, and performance that you can rely on. The following assurances reflect the standards, technologies, and practices that define how Poddar Pipes operates, and why professionals across the industry continue to choose us.

PRECISION MANUFACTURING LINES

Manufactured on state-of-the-art extrusion lines that are designed and maintained to deliver consistent product quality, precise dimensional accuracy, and dependable long-term performance across every single unit produced.

100% VIRGIN RAW MATERIALS

Made exclusively using 100% virgin raw materials, ensuring that every pipe and fitting delivers superior structural strength, enhanced durability, and a level of product consistency that recycled or inferior materials simply cannot match.

INTERNATIONAL QUALITY STANDARDS

All pipes and fittings are manufactured in strict compliance with IS 16098 and EN 1401 standards, providing customers with the assurance that every product meets nationally and internationally recognised quality benchmarks.

BATCH-BY-BATCH DIMENSIONAL VERIFICATION

Every production batch undergoes thorough dimensional checks and verification procedures before dispatch, ensuring that all products conform precisely to the required specifications and applicable standards before reaching the customer.

TESTED FOR UNDERGROUND STRENGTH

Ring stiffness is comprehensively tested on all relevant products to confirm reliable underground performance, resistance to external loading, and the long-term structural stability required for subsurface installations.

LIGHTWEIGHT YET STRUCTURALLY RIGID

Advanced foam core technology is employed to enhance overall pipe rigidity and structural integrity, while simultaneously reducing the total weight of the pipe for significantly easier on-site handling and faster installation.

GUARANTEED LEAK-PROOF JOINTS

Fully leak-proof joints are consistently achieved through the use of pre-fitted Brown Seal™ rubber rings, which are engineered to deliver a secure, dependable, and long-lasting sealing performance across all connection points.

ISI CERTIFIED, INDUSTRY COMPLIANT

All ISI certified products are manufactured in full compliance with applicable government and industry standards, ensuring that every product meets the required benchmarks for quality, operational safety, and long-term reliability.

ITALIAN TECHNOLOGY, GLOBAL EXPERTISE

Manufactured using advanced European manufacturing technology and backed by decades of proven industry expertise, our products are engineered to bring world-class piping solutions and global standards to the Indian market.

VERSATILE RANGE, ENDLESS APPLICATIONS

Available across a comprehensive range of sizes and compatible fittings, our products are designed to meet the diverse requirements of both residential and commercial applications with equal ease and reliability.

BUILT DIFFERENT, BUILT BETTER

STRENGTH, SAFETY AND **SCIENCE OF uPVC**

Unplasticized polyvinyl chloride, commonly known as uPVC, combines the structural benefits of polyvinyl chloride with the rigidity of an unplasticized form. Pipes manufactured from uPVC are exceptionally durable and built to last for decades. Unlike metal pipes, uPVC does not rust, rot, or deteriorate over time, making it a highly reliable choice for cold water applications including plumbing, water supply networks, underground drainage, and sewage lines.

PHYSICAL PROPERTIES

One of the standout characteristics of uPVC pipe is its ability to tolerate significant ground movement and structural bending without fracturing or losing integrity. This makes it an increasingly preferred piping material in earthquake-prone regions, where pipelines must endure repeated ground vibrations without sustaining damage or compromising the water supply.

The smooth interior surface of uPVC pipe improves flow efficiency while providing natural resistance to bacterial contamination, including harmful organisms such as E. coli, making it a trusted material for water distribution companies and municipal bodies requiring contamination-free pipelines.

Physical Property	Value
Density [g/cm^3]	1.300 - 1.450
Thermal conductivity [w/(m.k)]	0.14 - 0.28
Yield strength [MPa]	31 - 60
Young's modulus [psi]	490,000
Flexural strength (yield) [psi]	10,500
Compression strength [psi]	9500
Coefficient of thermal expansion (linear) [mm(mmoC)]	5×10^{-5}
Vicat B [°C]	>80°C
Resistivity [Ωm]	10^{16}
Surface resistivity [Ω]	10^{13} - 10^{14}

FIRE RESISTANT

Poddar Agri pipes and fittings are self-extinguishing and do not support combustion, making them well-suited for residential and commercial projects where fire safety is a priority. uPVC requires sustained direct flame exposure to ignite, owing to its high Limiting Oxygen Index (LOI) of 45.

The LOI measures the minimum oxygen percentage required to sustain combustion. Since the Earth's atmosphere contains only 21% oxygen, these pipes will not ignite under normal conditions and will cease burning immediately once the ignition source is removed.

Material	LOI
Cotton	16 - 17
Polypropylene (PP)	18
Polyethylene (PE)	18
Wood	20
Atmospheric content of OXYGEN	21
uPVC	45
CPVC	60

THE PODDAR AGRI DIFFERENCE

WHY PROFESSIONALS RELY ON PODDAR AGRI

Poddar Pipes manufactures a Lead Free uPVC Agri solvent weld system, purpose-built as a reliable solution for agricultural water supply. Agri pipes are manufactured in compliance with IS 4985:2000 standards and are available in sizes ranging from 40 mm to 315 mm across different pressure classes. Fittings are available from 40 mm to 110 mm.

LEAD FREE FORMULA

Poddar Agri is a completely lead free system, making it a safe and ideal choice for potable water distribution. It conforms to the latest international standards for drinking water supply pipes and is widely preferred worldwide.

ZERO TOXICITY, ODOUR AND TASTE

Poddar Agri pipes and fittings are entirely non-toxic, odourless, and tasteless. They are designed to preserve the natural quality and purity of the water being conveyed, making them fully suitable for potable water applications.

BUILT STRONG, BUILT TO LAST

Poddar Agri pipes and fittings are highly resilient, tough, and engineered for long-term use. They offer high tensile and impact strength, withstand sustained high pressure, and remain free from corrosion, rust, weathering, and chemical degradation throughout their service life.

COMPLETELY MAINTENANCE FREE

Poddar Agri pipes and fittings do not rust, pit, scale, or corrode over time. Once correctly installed, they deliver years of uninterrupted, trouble-free service with no ongoing maintenance required.

EFFORTLESS INSTALLATION

Poddar Agri pipes and fittings are lightweight and feature smooth, seamless interior walls. They require no specialist tools for cutting and are joined simply and effectively using solvent cement, making the entire installation process straightforward and efficient.

COMPREHENSIVE CORROSION RESISTANCE

Internal: Poddar Agri pipes resist chemical attack from most acids, alkalis, salts, and organic media including alcohols and aliphatic hydrocarbons, within defined temperature and pressure limits.
External: Industrial fumes, humidity, saltwater, and underground conditions regardless of soil type pose no threat. Surface scratches do not create points of vulnerability for corrosive attack.

SIMPLE AND RELIABLE JOINTING

The jointing process for Poddar Agri pipes is a straightforward single-step operation using solvent cement. This simple method consistently delivers a fully secure and 100% leak proof joint every time without the need for complex tools or techniques.

COST EFFICIENT SOLUTION

Poddar Agri pipes and fittings are lightweight, directly reducing transportation and handling costs on site. Their efficient installation process lowers labour time and overall project expenditure, making them a highly cost-effective piping solution from procurement to completion.

UV DEGRADATION RESISTANCE

Poddar Agri pipes and fittings are manufactured with built-in UV resistance, protecting them against the degrading effects of prolonged ultraviolet exposure and ensuring structural integrity and performance are maintained even in direct outdoor sunlight.

RELIABLE CHEMICAL RESISTANCE

Poddar Agri pipes and fittings are chemically inert and resistant to a wide range of strong acids, alkalis, salt solutions, and alcohols. They do not react with the media being conveyed and do not act as a catalyst for any chemical process.

MINIMAL THERMAL CONDUCTIVITY

Poddar Agri pipes and fittings have a significantly lower thermal conductivity factor compared to metal pipes. This ensures that fluids flowing through the system maintain a consistent and stable temperature throughout the entire length of the pipeline.

FIRE SAFETY ASSURED

Poddar Agri pipes and fittings are self-extinguishing and do not actively support combustion. This property significantly enhances on-site and in-use safety by effectively limiting the spread of flames in the event of accidental or unexpected fire exposure.

TESTED AND PROVEN PROPERTIES

PODDAR AGRI PIPES **PROPERTIES**

GENERAL

Physical properties of uPVC pipe	Value	Test Method
Cell classification	12454	ASTMD1784
Maximum service temperature	140°F/60°C	-
Colour	Grey	-
Water Absorption % increase 24 hrs @ 25°C	0.05	ASTM D570
Rockwell hardness	110-120	ASTM D785
Poisson's Ratio @ 73°F	0.410	-
Hazen Williams factor	C=150	-

MECHANICAL

Physical properties of uPVC pipe	Value	Test Method
Specific gravity	1.400 / 1.460	ASTM D792
Tensile strength, psi @ 73°F	7,000	ASTM D638
Modulus of elasticity, psi @ 73°F(tensile modulus)	420,000	ASTM D638
Flexural strength, psi @ 73°F	14,450	ASTM D790
Compressive strength, psi @ 73°F	9,600	ASTM D695
Izod impact, ft-lb./in @ 73oF	0.75	ASTM D256

THERMAL

Physical properties of uPVC pipeivv	Value	Test Method
Coefficient of linear expansion (in/in/°F)	2.9 x 10-5	ASTM D696
Coefficient of thermal conductivity (BTU/hr/ft2/°F/in)	1.02	ASTM C177
Heat deflection temperature °F @ 264 psi	170	ASTM D648
Specific heat, Cal/g/°C	0.25	ASTM D2766

ELECTRICAL

Physical properties of uPVC pipe	Value	Test Method
Dielectric strength, V/mil	1,413	ASTM D149
Dielectric Constant, 60Hz, 30°F	3.7	ASTM D150
Volume resistivity, Ω/cm @ 95°C, ohms/cm	1.2×10^{12}	ASTM D257

FLAMMABILITY

Physical properties of uPVC pipe	Value	Test Method
Flammability rating	V-0	UL94
Flammability index	<10	-
Flame spread	0-25	ULC S102.2
Flash ignition temperature	730°F	ASTM D1929
Average time of burning (sec.)	<5	ASTM D635
Average extent of burning	<10(mm)	ASTM D635
Burning rate (in/min)	Self Extinguishing	ASTM D635
Softening starts (approx)	250°F/121°C	-
Material becomes viscous	350°F/176°C	-
Material carbonizes	425°F/218°C	-
Smoke generation	80-225	ULC S102.2

STANDARDS FOR PIPES AND FITTINGS

Class of Pipe/Fitting	Standard	Sizes Available
Class 2 (4 Kgf/cm2) Pipe	IS 4985	63 mm - 315 mm
Class 3 (6 Kgf/cm2) Pipe	IS 4985	40 mm - 315 mm
Class 3 (6 Kgf/cm2) Fitting	IS 7834	63 mm - 110 mm
Class 5 (10 Kgf/cm2) Fitting	IS 7834	40 mm - 50 mm

PVC PIPES TEMPERATURE DE-RATING FACTOR FOR PRESSURE RATING

Operating Temperature		
Fahrenheit (°F)	Centigrade (°C)	De-rating Factor
73	23	1.00
80	27	0.88
90	32	0.75
100	38	0.62
110	43	0.51
120	49	0.40
130	54	0.31
140	60	0.22

SIZES WE MANUFACTURE IN

PODDAR AGRI PIPES
DIMENSIONS

DIMENSIONS AS PER IS 4985 - CLASS 2 (4KGF/CM2)

Nominal Outside Diameter (Nominal Size- mm)	Mean Outside Diameter (O.D)		Wall Thickness	
	Minimum (mm)	Maximum (mm)	Minimum (mm)	Maximum (mm)
63	63.00	63.30	1.50	1.90
75	75.00	75.30	1.80	2.20
90	90.00	90.30	2.10	2.60
110	110.00	110.40	2.50	3.00
125	125.00	125.40	2.90	3.40
140	140.00	140.50	3.20	3.80
160	160.00	160.50	3.70	4.30
180	180.00	180.60	4.20	4.90
200	200.00	200.60	4.60	5.30
225	225.00	225.70	5.20	6.00
250	250.00	250.80	5.70	6.50
315	315.00	316.00	7.20	8.30

DIMENSIONS AS PER IS 4985 - CLASS 3 (6KGF/CM2)

Nominal Outside Diameter (Nominal Size- mm)	Mean Outside Diameter (O.D)		Wall Thickness	
	Minimum (mm)	Maximum (mm)	Minimum (mm)	Maximum (mm)
40	40.00	40.30	1.40	1.80
50	50.00	50.30	1.70	2.10
63	63.00	63.30	2.20	2.70
75	75.00	75.30	2.60	3.10
90	90.00	90.30	3.10	3.70
110	110.00	110.40	3.70	4.30
125	125.00	125.40	4.30	5.00
140	140.00	140.50	4.80	5.50
160	160.00	160.50	5.40	6.20
180	180.00	180.60	6.10	7.10
200	200.00	200.60	6.80	7.90
225	225.00	225.70	7.60	8.80
250	250.00	250.80	8.50	9.80
315	315.00	316.00	10.70	12.40

MANUFACTURING STANDARDS

OUR STRINGENT QUALITY **CONTROL PROCEDURES**

Every pipe and fitting manufactured at Poddar Pipes undergoes a comprehensive and stringent quality control process before it is released into the market. Each stage of production is carefully monitored and evaluated against the highest applicable standards to ensure that only defect-free, fully compliant products reach our customers and perform reliably throughout their operational lifetime.

PIPES

EFFECT ON WATER

Each sample is tested to confirm that the pipe does not alter the taste, odour, or chemical composition of the water passing through it, ensuring clean, safe, and uncontaminated delivery from source to end user.

HEAT REVERSION TEST

This test measures how much the pipe changes in length when heated in an oven and subsequently left to cool.
It determines the level of residual stresses retained during production, confirming that pipes remain dimensionally stable under real-world temperature fluctuations throughout their service life.

HYDROSTATIC PRESSURE TEST

Short Term (Acceptance Test) at 20°C: When subjected to internal hydrostatic pressure, the pipe must not burst or crack at the given test pressure for a minimum of 1 hour, at a pressure exceeding 3 times the normal pressure rating.
Long Term (Type Test) at 95°C: The pipe must not crack or burst at the given test pressure for a period of 165 hours or 1,000 hours, validating its ability to perform reliably under sustained high-pressure conditions over time.
Thermal Stability at 95°C: The pipe shall not fail at the prescribed test pressure for a period of 8,760 hours, confirming that Poddar CPVC pipes maintain their structural integrity through an entire year of continuous high-temperature operation.

DROP IMPACT TEST

Weights are dropped onto the pipe to observe for any cracks or failures, simulating the physical stress a pipe may face during transportation, handling, and installation. Only pipes that show zero cracking or structural failure under impact are cleared for the next stage, ensuring that what reaches the site is built to handle real-world conditions without compromise.

FLATTENING TEST

Samples are compressed until opposite walls meet without the pipe cracking, confirming uniform wall thickness and correct extrusion during production.
A pipe that passes this test demonstrates consistent material distribution throughout its entire length, serving as a direct indicator of manufacturing precision and long-term durability under external load.

TENSILE STRENGTH

The maximum stress a pipe can withstand while being stretched or pulled, validating its structural integrity under the demands of installation and long-term use. Pipes meeting the required tensile strength threshold are confirmed to resist deformation and breakage even when subjected to mechanical force during handling or installation on site.

FITTINGS

STRESS RELIEF TEST

To determine internal stress levels by heating the fitting in an air-circulated oven at 150°C. There should be no blisters, weld line splitting, or cracking, ensuring every fitting is free from defects that could cause failure under pressure and confirming that each fitting forms a secure, leak-free joint throughout years of continuous use.

GUIDELINES FOR **HANDLING & STORAGE**

Every pipe, fitting, and solvent cement product requires proper care during handling and storage to ensure it reaches installation in perfect condition and performs as intended throughout its service life.

PROPER HANDLING OF PIPES

INSPECT ON ARRIVAL

Upon receiving your order, thoroughly inspect every pipe for any signs of damage that may have occurred during transit, including shifts in load or mishandling during transportation. Pay close attention to the ends of each pipe and examine them carefully for visible cracks, chips, or any structural damage before accepting the delivery or proceeding with installation.

HANDLE WITH CARE

Always handle pipes with the care and attention they deserve. Never throw, drop, or drag pipes from a truck bed or any elevated surface, as this can cause invisible stress fractures that compromise performance. Keep pipes away from sharp objects and edges at all times during movement, as any contact with such surfaces can result in damage that affects the long-term integrity of the pipe.

STORAGE OF PIPES

CHOOSE THE RIGHT LOCATION

Pipes should ideally be stored indoors in a covered and protected environment. When indoor storage is not available or practical, ensure that pipes are stacked no higher than the recommended limit and protected from the elements. Always store pipes on a level, dry surface and shield them from direct sunlight to minimise the effects of prolonged UV exposure on the material.

STACK AND SUPPORT CORRECTLY

When stacking pipes of the same diameter but belonging to different pressure classes, always position the thicker-walled pipes at the bottom of the stack to maintain structural stability. If pipes are being stored on racks, ensure that the spacing between supporting points does not exceed 3 feet to prevent bowing or deformation of the pipe over time.

SAFE HANDLING OF SOLVENT CEMENT

SEAL AFTER EVERY USE

After each and every application of solvent cement onto a pipe or fitting, immediately replace the lid on the container and tighten it securely to prevent evaporation and the unintended escape of solvent fumes into the surrounding environment.

ENSURE ADEQUATE VENTILATION

Avoid inhaling solvent vapours for any extended period of time. Whenever pipes and fittings are being joined in enclosed or confined areas, it is essential to ensure that adequate ventilation is maintained throughout the entire jointing process.

PROTECT EYES AND SKIN

Prevent direct contact of solvent cement with the eyes and skin at all times. If eye contact does occur, immediately flush the affected area thoroughly with a generous amount of clean water for a minimum of 15 minutes and seek prompt medical attention.

STORE CONTAINERS PROPERLY

All containers of solvent cement, primers, and cleaners must be kept tightly sealed and closed at all times except during active use. Any rags or applicator materials that have been used with solvents must be disposed of responsibly in a designated outdoor waste bin.

AWAY FROM IGNITION SOURCES

Solvent cement, primers, and cleaning agents must be stored and used well away from all potential sources of ignition, including open flames, heat sources, sparks, and any equipment that could trigger combustion in the presence of flammable vapours.

INSTRUCTIONS AND GUIDELINES **FOR SOLVENT CEMENT**

RECOMMENDATIONS

One Step Solvent Cement is the recommended product for joining all pipes and fittings in the Poddar Agri system.

SUMMARY

1. The following procedures must be clearly understood and carefully followed at all times:

- The surfaces to be joined must be softened and dissolved to a semi-fluid state before assembly begins.
- A sufficient amount of solvent cement must be applied to fully fill the gap between the pipe and the fitting.
- The pipe and fitting must be assembled and brought together while both surfaces are still wet and in a fluid condition.
- Joint strength develops gradually as the solvent cement dries. In the tight section of the joint, the surfaces will tend to fuse directly together, while in the looser section, the One-Step solvent cement will bond firmly to both surfaces.

2. A sufficient quantity of One-Step solvent cement must be applied to completely fill the gap present in the loose section of the joint (see Figure 2).
In addition to filling this gap, adequately applied layers of One-Step solvent cement will penetrate into both contact surfaces and must remain wet and workable until the joint is fully assembled.

3. If the One-Step solvent cement coatings on both the pipe and fitting surfaces are still wet and fluid at the time of assembly, they will naturally flow into each other and merge to form a single, unified solvent cement layer. Additionally, while the solvent cement remains wet, the material beneath the coated surfaces will still be in a softened state, and these dissolved surfaces within the tight section of the joint will tend to fuse together under the applied pressure (see Figure 3).

4. As the solvent gradually evaporates from the joint, the One-Step solvent cement layer and the dissolved surfaces beneath it will dry out and harden progressively, resulting in a corresponding and steady increase in overall joint strength. Completed joints must not be disturbed or subjected to any movement until they have cured sufficiently to withstand the stresses involved in normal handling and use. Joint strength continues to develop as the One-Step solvent cement fully dries and cures.

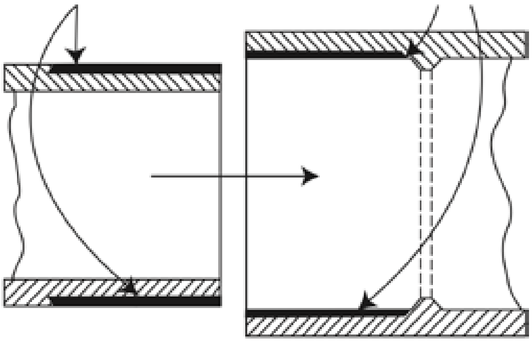


Figure 1: Outside of pipe and inside the fitting socket to be softened and penetrated

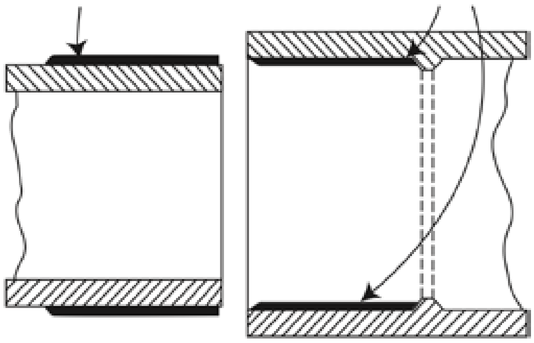


Figure 2: Solvent cement coatings of sufficient thickness applied uniformly around pipe and inside fitting socket.

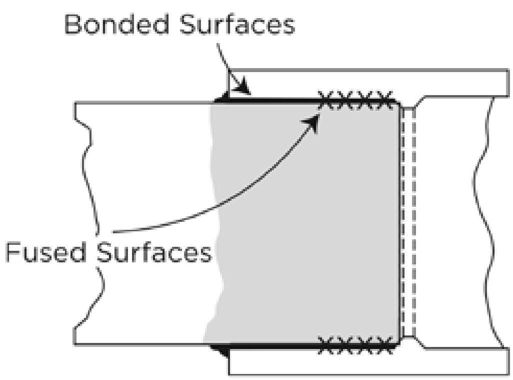


Figure 3: Fused and bonded surfaces of joined pipe and fitting.

WARNING

Follow all preparation and installation procedures without exception before use.

EASY AND 100% LEAKPROOF INSTALLATION

STEP-BY-STEP **INSTALLATION GUIDE**

STEP 1: CUTTING

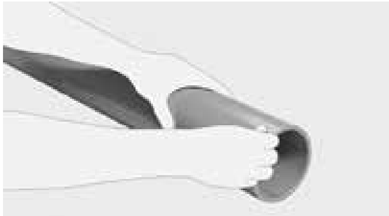
Measure the required pipe length and mark it clearly using a felt tip pen. Verify size compatibility before proceeding. Cut using a plywood saw, ratchet cutter, or wheel cutter, always at 90° to ensure an optimal bonding area. Inspect pipe ends thoroughly before jointing. If any crack or splintering is found, cut a minimum of 25 mm beyond the visible damage before proceeding.

STEP 2: DEBURRING / BEVELING

Burrs on or inside the pipe end can obstruct flow and prevent proper contact between pipe and fitting socket during assembly. Remove them carefully from both inner and outer surfaces using a 15 mm half-round file, pen knife, or deburring tool. Applying a slight bevel to the pipe end will ease its entry into the fitting socket during assembly.

STEP 3: FITTING PREPARATION

Wipe all dirt, dust, and moisture thoroughly from the fitting sockets and pipe end using a clean, dry rag to ensure a clean bonding surface. Perform a dry fit to confirm the pipe enters fully and reaches the bottom of the socket without obstruction. Once confirmed, mark the pipe clearly at the socket entry point using a felt tip pen for accurate reference during final assembly.

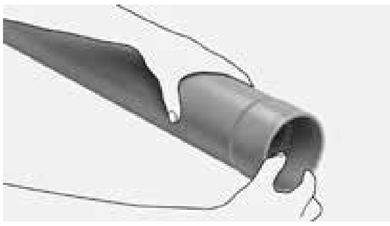


STEP 4: ONE STEP SOLVENT CEMENT PROCEDURE

Use only genuine Poddar uPVC Solvent Cement conforming to ASTM D-2564 / IS 14182 for a reliable solvent weld joint. Apply a smooth, even coat uniformly across the pipe end and inside the fitting socket. Never use thickened or lumpy cement. It must flow freely with a consistency similar to syrup or paint for proper surface penetration.

STEP 5: ASSEMBLY

Immediately after applying solvent cement, insert the pipe firmly into the fitting socket. Rotate the pipe a quarter to half turn while inserting to distribute the cement evenly across the joint surface. Hold the assembly firmly in position for a minimum of 3 seconds to allow the joint to begin setting and prevent push-out.



STEP 6: FINISHING

Check that a continuous bead of solvent cement has formed around the full circumference of the socket fitting entrance, confirming sufficient cement coverage and a fully sealed joint. Using a clean, dry cloth, wipe away excess solvent cement from the external surface of both the pipe and fitting for a clean, professional finish.

TEMPERATURE AND TIME GUIDELINES

SOLVENT CEMENT
SET & CURE TIMES

AVERAGE INITIAL SET SCHEDULE FOR uPVC SOLVENT CEMENT

Temperature Range	Pipe Sizes ½"-1¼" (15 mm – 32 mm)	Pipe Sizes 1½"-2" (40 mm – 50 mm)	Pipe Sizes 2½"-6" (65 mm – 150 mm)
60° – 100°F / 16° – 38°C	2 minutes	5 minutes	30 minutes
40° – 60°F / 5° – 16°C	5 minutes	10 minutes	2 hours
0° – 40°F / -18° – 5°	10 minutes	15 minutes	12 hours

Note - Initial set schedule is the necessary time to allow before the joint can be carefully handled. In damp or humid weather allow 50% more set time.

AVERAGE JOINT CURE SCHEDULE FOR uPVC SOLVENT CEMENT

Relative Humidity 60% or Less	Pipe Sizes ½"-1¼" (15 mm – 32 mm)		Pipe Sizes 1½"-2"" (15 mm – 32 mm)		Pipe Sizes 2½"-6" (15 mm – 32 mm)	
Temperature range during assembly and cure periods	psi (Bar)		psi (Bar)		psi (Bar)	
	up to 160 (up to 11)	160 to 370 (11 to 26)	up to 160 (up to 11)	160 to 315 (11 to 22)	up to 160 (up to 11)	160 to 315 (11 to 22)
60° – 100°F / 16° – 38°C	15 minutes	6 hours	30 minutes	12 hours	1 - ½ hours	24 hours
40° – 60°F / 5° – 16°C	20 minutes	12 hours	45 minutes	24 hours	4 hours	48 hours
0° – 40°F / -18° – 5°	30 minutes	48 hours	1 hour	96 hours	72 hours	8 days

Note - Joint cure schedule is the necessary time to allow before pressurizing system. In damp or humid weather allow 50% more cure time.

CAUTION

These figures are estimates based on testing done under laboratory conditions. Although this information is widely published across the industry, these charts should be used as a general reference only. Field working conditions can vary significantly and will increase set and cure times.

PRESSURING SOLVENT CEMENT JOINTS

Adequate time must always be allowed for solvent cement joints to fully cure and develop their complete rated strength before any pressure is introduced into the system. Several site-specific factors will directly influence the amount of cure time required before a joint can be safely pressurised. It is important to note that these factors can vary significantly from one installation to the next. In general, the cure times provided in the schedule above will allow cold water lines to be safely pressurised to the pressure levels indicated in the corresponding table.

These factors include:

a. On-site ambient temperature and relative humidity levels during installation.
b. Pipe diameter, as larger diameter joints inherently require a greater amount of time to cure fully.
c. The internal operating pressure that the system will ultimately be subjected to.
d. The internal operating temperature of the fluid being conveyed.

ESSENTIAL TIPS AND **INSTALLATION WARNINGS**

HOT WEATHER INSTALLATION TIPS

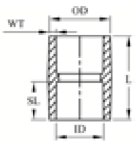
1. Store solvent cement in a cool or well-shaded location prior to use to maintain its consistency and effectiveness.
2. Where possible, store pipes and fittings in a shaded area before applying solvent cement to prevent surfaces from becoming too warm.
3. Cool down the surfaces to be joined using a clean, damp rag before beginning. Ensure the surface is completely dry before applying solvent cement.
4. Where possible, carry out solvent cement application during the cooler morning hours to allow better working time.
5. Always confirm that both surfaces being joined are still visibly wet with solvent cement at the moment they are brought together.
6. Vigorously stir or shake the solvent cement container thoroughly before each use to ensure a uniform consistency.
7. System anchoring and all final pipe connections should be completed during the cooler parts of the day to appropriately account for thermal expansion and contraction.

INSTALLATION WARNING

1. Always perform a complete dry fit of all joints prior to applying any solvent cement, to confirm that a proper and secure interference fit exists between components.
2. Any fitting joints that do not demonstrate a correct and adequate interference fit during dry fitting must be discarded and replaced with suitable alternatives immediately.
3. Do NOT apply solvent cement to any joint that feels either too loose or excessively tight, as improper fit will result in a weak or failed joint.
4. Always use appropriate and purpose-made bevelling tools to correctly prepare all pipe ends before any solvent cement is applied to ensure a proper bond.
5. Do NOT apply solvent cement to any joint without first completing the bevelling of all pipe ends, as unbevelled ends will compromise the joint quality.
6. Use only One-Step solvent cement when connecting all pipes, fittings, and associated accessories to ensure compatibility and a consistently strong, reliable, leak-proof joint throughout the system.
7. Do NOT use any primer in combination with One-Step solvent cement, as the use of primer is neither required nor compatible with this jointing system.
8. Do NOT use any other brand or type of solvent cement to connect Poddar Agri pipes, fittings, and accessories, as this may result in joint failure.
9. Carefully read and follow all instructions provided for the correct application of solvent cement that are supplied together with this product before beginning installation.
10. Poddar fully supports and endorses all safety and protective measures recommended by relevant government agencies and regulatory bodies when installing Poddar Agri pipes on site.
11. Always ensure that adequate and proper ventilation is maintained throughout the entire work area whenever solvent cement is being applied during the installation process.
12. Avoid all unnecessary contact of solvent cement with the skin or eyes during handling and application to prevent irritation or any adverse health effects.
13. In the event of any accidental skin or eye contact with solvent cement, wash the affected area immediately and thoroughly with clean water to prevent prolonged exposure.
14. Strictly follow all manufacturer-recommended safety precautions when cutting or sawing pipe, or when operating any flame, heat-generating, or power tools near the installation area.
15. Keep all open flames, soldering equipment, and sources of ignition well away from solvent cement joints at all times during and after the installation

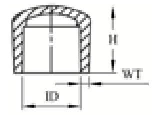
TECHNICAL DETAILS

PODDAR FITTINGS
DIMENSIONS

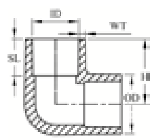
COUPLER

SIZE	ID	SL	WT	L
40	40.10 / 40.30	26.00	3.00	55.20
50	50.10 / 50.30	31.50	3.00	66.00
63	63.10 / 63.30	38.00	3.00	79.50
75	75.10 / 75.30	44.00	3.10	92.00
90	90.10 / 90.30	51.20	3.80	108.50
110	110.10 / 110.40	61.30	4.10	129.00

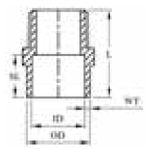
END CAP

SIZE	ID	SL	WT	L
40	40.10 / 40.30	26.20	3.20	34.30
50	50.10 / 50.30	31.70	3.20	41.30
63	63.10 / 63.30	38.00	3.20	47.30
75	75.10 / 75.30	44.00	3.20	52.00
90	90.10 / 90.30	51.00	3.20	62.30
110	110.10 / 110.40	61.00	3.20	69.50

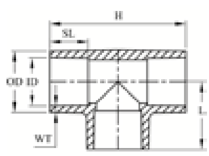
ELBOW 90°

SIZE	ID	SL	WT	H
40	40.10 / 40.30	26.00	3.00	72.30
50	50.10 / 50.30	31.50	3.00	87.60
63	63.10 / 63.30	38.00	3.00	106.90
75	75.10 / 75.30	44.00	3.10	123.80
90	90.10 / 90.30	51.20	3.10	145.60
110	110.10 / 110.40	61.30	3.30	177.50

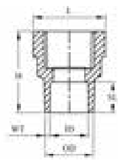
MALE ADAPTER PLASTIC THREADED (MAPT)

SIZE	ID	SL	WT	H
40	40.10 / 40.30	26.00	3.00	54.00
50	50.10 / 50.30	31.00	3.00	61.00
63	63.10 / 63.30	38.00	3.20	70.00
75	75.10 / 75.30	44.00	3.20	82.50
90	90.10 / 90.30	51.00	3.80	94.00
110	110.10 / 110.40	61.00	4.80	106.00

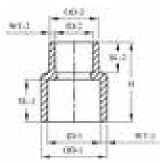
TEE

SIZE	ID	SL	WT	H
40	40.10 / 40.30	26.00	3.00	96.70
50	50.10 / 50.30	31.50	3.00	118.40
63	63.10 / 63.30	38.00	3.00	143.90
75	75.10 / 75.30	44.00	3.10	92.00
90	90.10 / 90.30	51.20	3.10	108.50
110	110.10 / 110.40	61.30	4.10	129.00

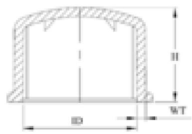
FEMALE ADAPTER PLASTIC THREADED (FAPT)

SIZE	ID	SL	WT	H
40	40.10 / 40.30	26.00	3.00	55.00
50	50.10 / 50.30	31.50	3.00	65.00
63	63.10 / 63.30	38.00	3.20	74.00
75	75.10 / 75.30	44.00	3.20	91.00
90	90.10 / 90.30	51.00	3.30	106.00
110	110.10 / 110.40	61.50	3.80	118.00



REDUCER COUPLER

SIZE	ID-1	SL-1	WT-1	ID-2	SL-2	WT-2	H
50 x 40	50.38	31.00	3.20	40.36	26.00	3.20	66.50
63 x 40	63.42	38.00	3.00	40.36	26.50	3.10	80.50
63 x 50	63.42	38.00	3.00	50.38	31.50	3.00	80.50
75 x 40	75.44	44.00	3.00	40.36	26.50	3.00	93.50
75 x 50	75.44	44.00	3.00	50.38	31.50	3.00	93.50
75 x 63	75.44	44.00	3.00	63.42	38.00	3.00	93.50
110 x 90	110.40	61.30	3.00	90.40	51.30	3.00	126.00

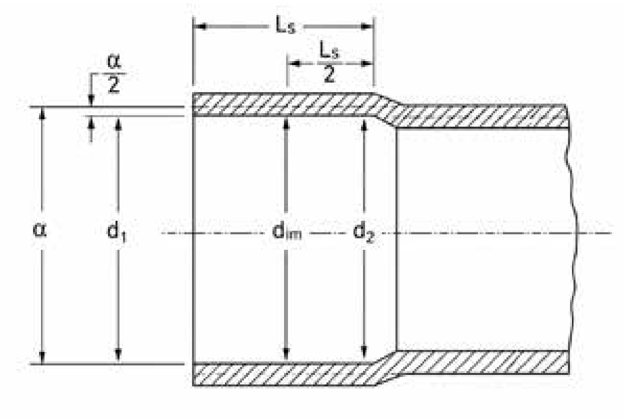


THREADED END CAP

SIZE	ID	WT	H
40	42.36	3.20	30.50
50	48.27	3.20	32.70
63	59.79	3.20	37.50
75	75.44	3.20	42.00
90	88.40	3.20	50.00
110	113.30	3.20	54.00

TECHNICAL DETAILS

PODDAR SOCKETS
DIMENSIONS



d_{im} = diameter at mid point of socket length
d_1 = diameter at mouth
d_2 = diameter at root
l_s = socket length
α = total included angle of taper

Nominal Size (DN)	Socket Length (L_s)	Mean Socket Internal Diameter at Mid-Point of Socket Lenght (d_{im})	
	Minimum	Minimum	Maximum
20	16.0	20.1	20.3
25	19.0	25.1	25.3
32	22.0	32.1	32.3
40	26.0	40.1	40.3
50	31.0	50.1	50.3
63	37.5	63.1	63.3
75	43.5	75.1	75.3
90	51.0	90.1	90.3
110	61.0	110.1	110.4
125	68.5	125.1	125.4
140	76.0	140.2	140.5
160	86.0	160.2	160.5
180	96.0	180.2	180.5
200	106.0	200.3	200.6
225	118.0	225.3	225.7
250	131.0	250.4	250.8
280	146.0	280.4	280.9
315	163.5	315.4	316.0
355	183.5	355.4	356.0
400	206.0	400.4	401.0
450	231.0	450.4	451.0
500	256.0	500.4	501.0
560	286.0	560.4	561.0
630	321.0	630.4	631.0

KNOW MORE

FREQUENTLY ASKED **QUESTIONS**

FAQs

WHY LEAD FREE?

A lead free water supply system is the safest and most preferred choice for potable water transportation worldwide. Eliminating lead from the pipe compound ensures that water conveyed remains completely uncontaminated and safe for human consumption.

WHY UV RESISTANT?

UV resistance prevents the oxidation process that occurs upon prolonged sunlight exposure, protecting the material from degradation and significantly extending the durability and service life of pipes and fittings in outdoor environments.

WHY CHOOSE PODDAR AGRI uPVC PIPES?

Poddar Agri uPVC pipes and fittings are produced using a specially formulated lead free compound that ensures full conformity to the most current requirements followed in developed nations. This lead free compound is completely non-toxic and certified safe for drinking water applications. All Poddar Agri pipes and fittings are manufactured under stringent quality control and dimensional accuracy standards, maintaining minimal ovality at the highest levels of precision. Poddar Agri fittings are produced using specially engineered moulds in which the gate point is deliberately positioned away from the weld line, significantly minimising the possibility of weld line failure under pressure.

WHAT ABOUT HEALTH SAFETY AND FIRE TOXICITY?

Independent testing conducted at multiple accredited laboratories confirms that uPVC performs better than metal-based piping systems in terms of its effects on water quality. In fire conditions, uPVC has been found to be no more toxic than wood. The Poddar Agri uPVC system is manufactured from a fully lead free compound, making it the most preferred choice from a health and safety standpoint. With a Limiting Oxygen Index of 45, uPVC is effectively non-combustible under normal atmospheric conditions. As soon as the external ignition source is removed, the material ceases to burn immediately.

PRODUCT WARRANTY

LIMITED WARRANTY

Poddar Pipes Pvt. Ltd. warrants to the original purchaser and owner that all products supplied will remain free from manufacturing defects and will conform fully to the applicable and currently valid BIS standards when used under normal operating conditions. In the event of a valid warranty claim, the buyer's remedy is strictly limited to the replacement of, or a credit equivalent to the value of, the defective product in question. This warranty expressly excludes any costs associated with the removal, reinstallation, or replacement labour of any defective product, as well as any incidental, consequential, indirect, or punitive damages of any nature that may arise as a result of the defect.

THE LIMITED WARRANTY WILL NOT APPLY IF

1. Poddar products are used in combination with pipes, fittings, or solvent cement sourced from any other brand or manufacturer not approved by Poddar.
2. The product is used for any application or purpose other than the distribution and conveyance of domestic water within a standard residential or agricultural system.
3. The product experiences failure as a direct result of defects, errors, or deficiencies in the design, engineering planning, or execution of the installation work.
4. The completed joints are not subjected to a pressure test and verified as leak-free prior to the plastering or encasing of the pipe system within walls or structures.
5. The installation has not been carried out in full accordance with the guidelines and procedures set out in the official Poddar installation manual for this product.
6. The pipe sustains any form of mechanical damage caused by external forces such as nails, drilling activity, chiselling, or any other physical interference after installation.

SAME
PODDAR FAMILY
45 YEARS
OF TRUST

PODDAR
PIPES

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